

Database Schema Specification

Document Type: Technical Data Definition | Target Object: Service Request

1. Overview & Objectives

This specification establishes the structural data model for persisting AI-augmented Service Requests within the centralized database platform. The architecture details the collection of data logs that track user interactions alongside AI suggestions. This foundational schema enables end-to-end auditability, analytics on AI recommendation accuracy, and performance tuning of predictive categorization algorithms.

Design Note: Relational configurations must define `selectedCategoryId` and `aiSuggestedCategoryId` as relational foreign keys connecting directly to the global master categories dataset.

2. Field Specifications & Data Dictionary

Field Name	Data Type	Nullable	Functional Description
id	UUID / BIGINT	No	Unique system-generated primary identifier for the specific service request transaction record.
originalDescription	TEXT	No	The raw, unedited text description initially input by the client. Contains ambiguous or brief summaries.
improvedDescription	TEXT	Yes	The refined, high-clarity text block generated by the AI agent and officially confirmed/accepted by the customer.
selectedCategoryId	VARCHAR / INT	No	The operational reference ID of the finalized service classification category selected explicitly by the user.
aiSuggestedCategoryId	VARCHAR / INT	Yes	The distinct diagnostic category identifier automatically pre-selected or predicted by the AI engine.
usedAiCategorySuggestion	BOOLEAN	No	Audit flag capturing user conversion metric: <code>true</code> if the customer opted to apply the AI category choice; otherwise <code>false</code> .
usedAiDescriptionGenerator	BOOLEAN	No	Audit flag capturing feature engagement: <code>true</code> if the client engaged the AI text enhancement tool; otherwise <code>false</code> .
createdAt	TIMESTAMP	No	System date-time tracking entry creation, defaulted to standard UTC standard ISO notation.

3. Implementation Payload Model (JSON Schema Blueprint)

The following structural model maps standard API payload parameters utilized across frontend endpoints, middleware verification layers, and database input serialization handlers:

```
{
  "id": "req_987654321",
  "originalDescription": "pipe leaking in kitchen",
  "improvedDescription": "The main water pipe underneath the kitchen sink has a
```

```
steady leak, causing minor flooding in the cabinet.",  
  "selectedCategoryId": "cat_plumbing_01",  
  "aiSuggestedCategoryId": "cat_plumbing_01",  
  "usedAiCategorySuggestion": true,  
  "usedAiDescriptionGenerator": true,  
  "createdAt": "2026-07-09T12:45:00Z"  
}
```